

THE UNIVERSITY OF MELBOURNE

Final Report

COMP90072: The Art of Scientific Computation

Estimating Respiratory Frequency from ECG

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Study scope
This project uses the first hour of one ECG recording to compare two ways of estimating respiratory frequency. The results are compared with an airflow signal recorded at the same time.

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Abstract

Breathing can affect the timing and amplitude of ECG beats. This project compares two features, RR intervals and R–S amplitude, using the first hour of Apnea-ECG record a01. The hour was divided into 30 two-minute windows. Each feature was placed on a regular time grid, and its strongest frequency within a selected range was compared with the strongest frequency in the airflow signal.

In the main range of 0.15–0.40 Hz, the mean absolute error was 3.21 cycles/min for RR intervals and 1.76 cycles/min for R–S amplitude. Both errors increased when a wider range was used. R–S amplitude therefore agreed more closely with airflow in this recording. The study also found possible beat-annotation errors and missing respiratory samples. These issues, together with the use of only one recording, limit the conclusions. The analysis compares frequency estimates; it does not verify a clinical breathing-rate monitor.

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1 Introduction and data

Project question

An ECG records the electrical activity of the heart. Its R peaks provide a clear marker for the timing of each beat. Breathing can slightly change the time between beats, an effect called respiratory sinus arrhythmia [9]. Breathing movements can also change the size of the ECG waveform [3]. These changes may contain information about breathing frequency.

The project asks whether the strongest frequency in either of these ECG features agrees with an independently recorded respiratory signal. The main tasks are to read the data, identify the beat features, calculate their frequency spectra and compare the results. A spectrum shows how strongly different frequencies contribute to a signal.

Dataset

Record `a01` comes from the Apnea-ECG database [1, 2]. The ECG values are stored in `a01.dat`, with recording information in a01.hea. The file `a01.qrs` contains supplied heartbeat marks, while `a01.apn` contains one apnea label for each annotated minute.

Table 1: Main properties of the recording.

Property	Value
ECG channels and sampling rate	One channel at 100 Hz
Full ECG length	2,957,000 samples; 8 h 12 min 50 s
Segment analysed for respiration	First 60 minutes; 30 two-minute windows
Supplied heartbeat marks	29,938 across the full ECG
Minute labels across the full record	470 A labels and 19 N labels
Primary respiratory reference	Oronasal airflow from the paired `a01r` record

The paired `a01r` record also includes chest effort, abdominal effort and oxygen saturation. Airflow was used as the main respiratory reference; oxygen saturation was not used for frequency estimation. Both ECG and respiratory records cover the first hour.

Scope

The first-hour limit was a practical choice because downloading the full respiratory file was slow. The first two-minute window was used while developing the method, and the same settings were then used for the remaining windows. This is an exploratory study of one recording, rather than a test of performance across different people.

2 Reading and checking the ECG

The ECG was read using the format and calibration information in its header [4]. The sampling rate is 100 Hz, meaning that 100 samples were recorded each second. Sample positions were divided by 100 to obtain time in seconds. The gain of 200 and baseline of zero were used to convert stored ECG values into millivolts.

The WFDB tools were used to read the annotations [5, 6]. Plotting the heartbeat marks on the ECG allows their locations to be checked against the visible peaks. The upper panel in Figure 1 shows the first ten seconds, while the lower panel shows the minute labels across the full record.

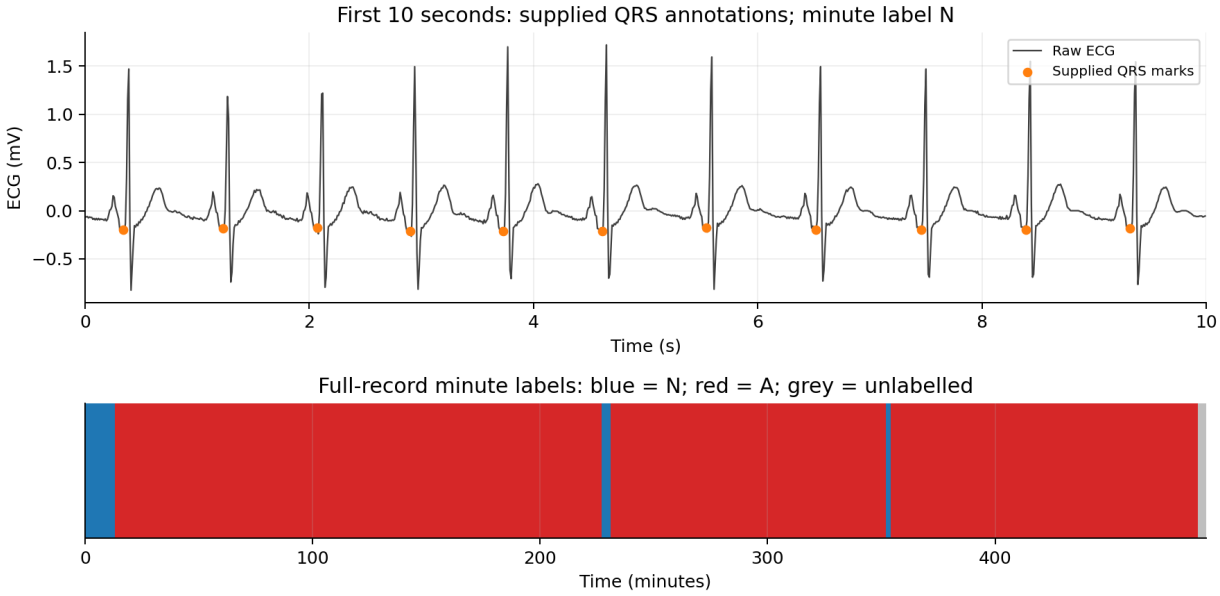


Figure 1: ECG and annotation checks. The upper panel shows 11 supplied heartbeat marks. The lower panel shows full-record minute labels; grey marks the unlabelled tail.

Understanding the annotations

The symbol N has different meanings in the two annotation files. In the QRS file it is a heartbeat symbol and does not confirm that every beat is normal. In the apnea file, N labels a minute without annotated apnea, while A labels a minute with apnea. An A label does not mean that breathing stopped for the whole minute.

The first-hour segment is fully covered by minute labels. The supplied heartbeat marks still require checking because the database describes them as unaudited [2].

3 Extracting the ECG features

Locating the R peaks

Some supplied heartbeat marks were slightly before the visible R peaks. Each mark was moved to the largest ECG value within 80 ms on either side. The shifts were 30–50 ms, with a median of 40 ms. This works with the upward R peaks in this recording, but cannot find a beat that has no supplied mark.

RR intervals and R–S amplitude

The RR interval is the time between two consecutive R peaks. If t_i is the time of the current R peak, then

$$RR_i = t_i - t_{i-1} \tag{1}$$

The second feature measures the vertical difference between an R peak and the lowest ECG value within the following 80 ms. This follows the feature definition in [3]. If R_i and S_i are those ECG amplitudes, then

$$A_i = R_i - S_i \tag{2}$$

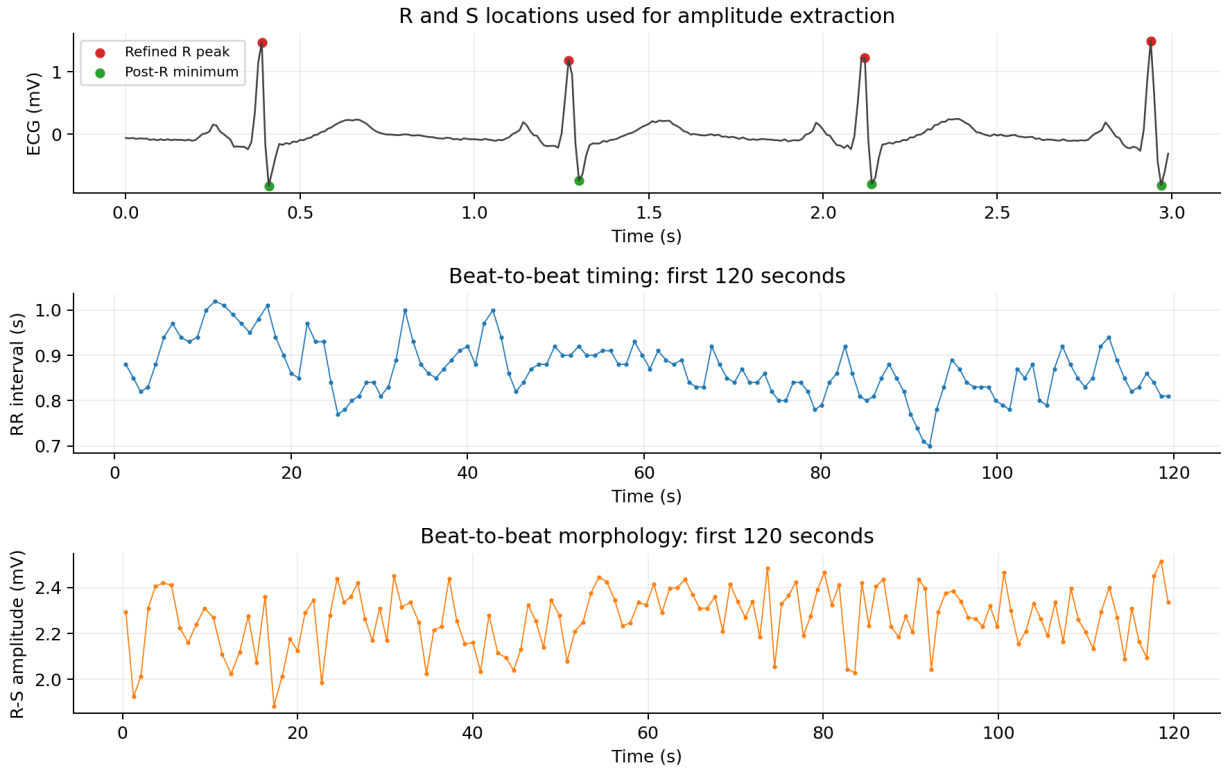


Figure 2: R peaks and the following minima, with the resulting RR intervals and R–S amplitudes. RR is measured in seconds and amplitude in millivolts.

4 Checking unusual RR intervals

An unusually long or short RR interval may indicate a beat-annotation problem. Each interval was compared with the median of up to five intervals on either side. A difference greater than 20% was flagged for inspection. This simple rule is a warning flag, not a confirmed diagnosis of an incorrect beat.

Across the full ECG, 443 of 29,937 intervals were flagged (1.48%). These full-record counts are separate from the 30-window first-hour comparison. Flagged intervals were kept in the main analysis so that difficult windows were not hidden.

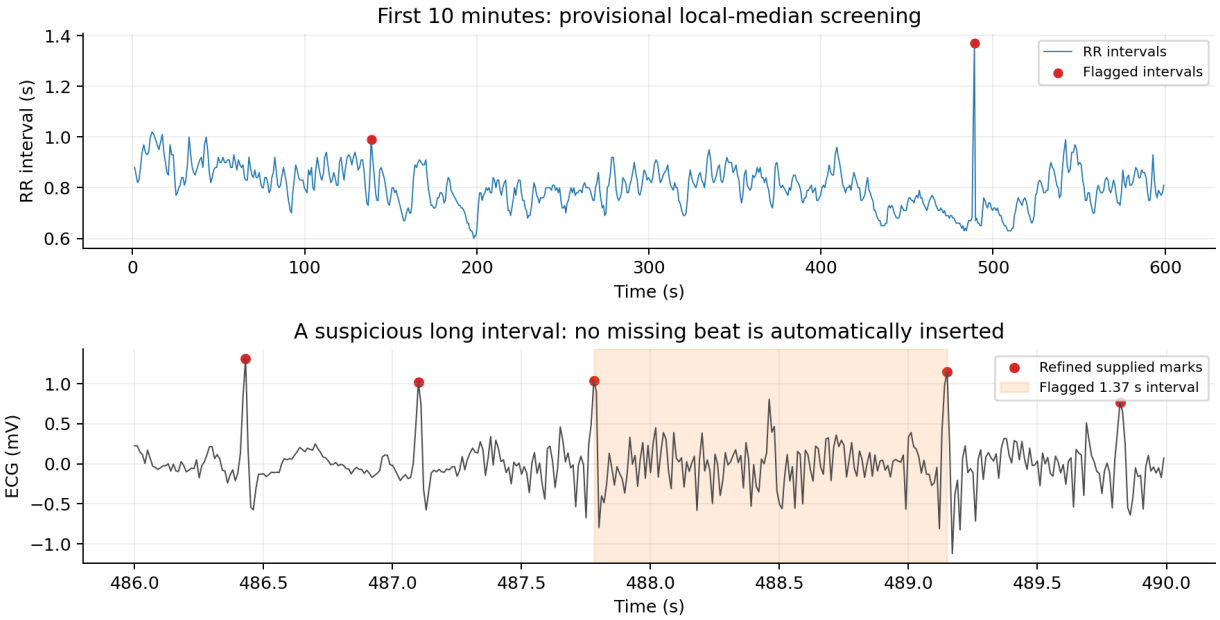


Figure 3: Flagged intervals in the first ten minutes. The lower panel shows the ECG around a long interval from 487.78 to 489.15 s.

Example of a possible missed beat

The interval shown in Figure 3 is 1.37 s, about twice its local median of 0.67 s. A missed beat is one possible explanation. However, the interval length alone does not show where that beat should be placed. It was therefore flagged and left unchanged, rather than adding a beat without enough evidence.

This decision avoids introducing an assumed beat, but leaves a possible error in the RR series. Some differences from airflow may therefore be caused by the heartbeat annotations.

5 Preparing the signals and estimating frequency

Using a common time grid

The first hour was divided into 30 non-overlapping two-minute windows. RR values occur at beat times, which are not equally spaced. Both ECG features were therefore linearly interpolated to 4 Hz, giving one value every 0.25 s. This creates a regular grid for frequency analysis; it does not add new measured beats. Only RR intervals with both beats inside a window were included.

The respiratory signals were also reduced from 100 to 4 Hz. The `resample_poly` function applies a low-pass filter before reducing the sampling rate, helping prevent high-frequency content from appearing at misleading lower frequencies [8]. The reference signals were then placed on the same time grid as the ECG features.

Missing respiratory samples

The first hour contained five missing chest samples and 80 missing abdominal samples. Airflow had none. Gaps were temporarily filled so filtering could run, but a channel estimate was rejected if its analysis interval, or a three-second margin on either side, contained an original gap. This affected one chest window and 22 abdominal windows. The airflow comparisons remained available for all 30 windows.

Finding the strongest frequency

A straight-line trend was removed from each signal to reduce slow drift. A Hann window was then applied to reduce abrupt changes at the segment ends. A periodogram, calculated with SciPy, estimated the power spectral density (PSD), showing how power is distributed across frequencies [7]. This calculation uses a fast Fourier transform (FFT) to analyse the signal by frequency. The frequency with the greatest power inside the search range was selected.

The main search range was 0.15–0.40 Hz. A wider range of 0.10–0.50 Hz was also tested to see how much the choice affected the answer. The selected frequency was converted from cycles per second to cycles per minute:

$$\text{Estimated rate} = 60 \times f_{\text{peak}} \tag{3}$$

For example, 0.20 Hz corresponds to 12 cycles/min. The report uses cycles/min because these are spectral estimates rather than manually counted breaths. A strong peak can also come from noise or other signal changes.

Comparing the estimates

Mean absolute error (MAE) is the average size of the difference between an ECG estimate and the airflow estimate. Root mean squared error (RMSE) gives more weight to large differences. Bias is the average signed difference; a negative value means that the ECG estimates tend to be lower. Here, all three describe agreement with the airflow spectrum, whose peaks have not been checked against individual breaths.

6 Results

The first two minutes

In the main frequency range, the RR estimate and airflow estimate were both 12.18 cycles/min. R–S amplitude gave 11.16 cycles/min. The airflow peak corresponds to a dominant period of about 4.93 s, although the waveform shows that individual cycles vary.

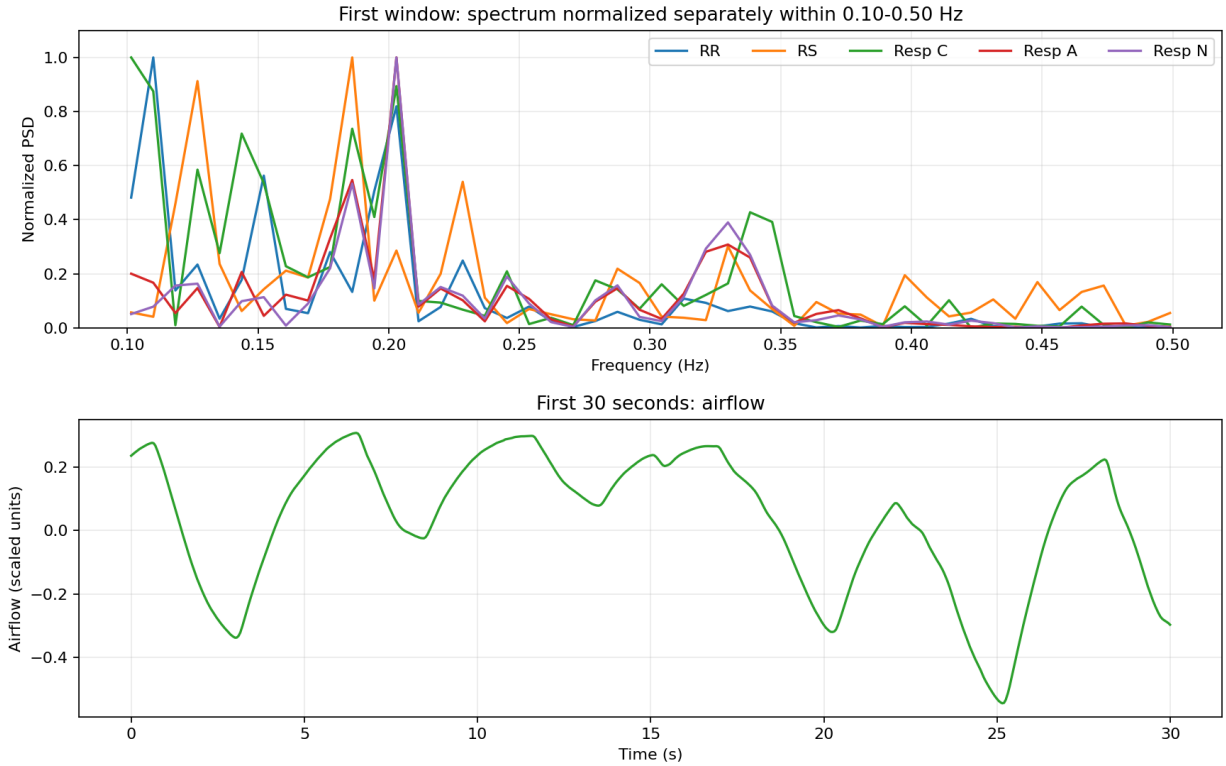


Figure 4: Spectra for the first window and the first 30 seconds of airflow. Resp C, A and N mean chest effort, abdominal effort and airflow. Each spectrum is divided by its own maximum, so heights show relative rather than absolute power.

Table 2: First-window frequency estimates, in cycles/min.

Signal	0.15–0.40 Hz	0.10–0.50 Hz
RR intervals	12.18	6.60
R–S amplitude	11.16	11.16
Chest effort	12.18	6.09
Abdominal effort	12.18	12.18
Airflow	12.18	12.18

The wider range changed the RR estimate to 6.60 cycles/min, while airflow stayed at 12.18. This shows that a good match in one window and one frequency range is not enough to judge the method. The comparison was therefore extended to all 30 windows.

Comparison across the first hour

All 30 windows had usable airflow and enough ECG features for analysis. The main results include windows with RR flags, missing abdominal data and large differences from airflow.

Table 3: Differences from the airflow estimate across all 30 windows. Errors and bias are in cycles/min.

Range in Hz	Method	MAE	RMSE	Bias
0.15–0.40	RR	3.21	4.94	−2.47
0.15–0.40	R–S	1.76	2.61	−0.81
0.10–0.50	RR	6.41	8.44	−4.52
0.10–0.50	R–S	2.33	3.82	0.10

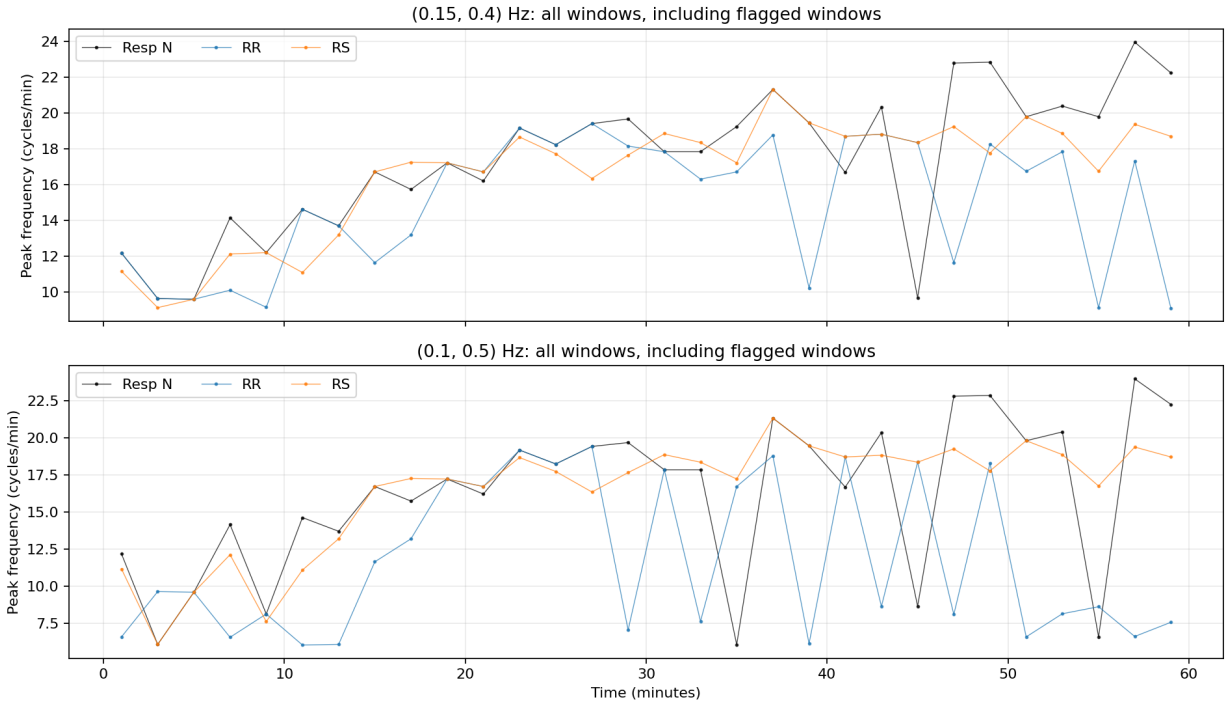


Figure 5: Frequency estimates across the first hour for the two search ranges. Lines connect the window estimates and do not show a continuously measured breathing rate.

R–S amplitude had lower MAE and RMSE in both ranges. In the main range, its MAE was 1.76 cycles/min compared with 3.21 for RR. Figure 5 shows several windows where the RR estimate drops well below airflow, contributing to the larger error.

Both methods had larger errors in the wider range. The airflow search range also changed, so these results show sensitivity to the range rather than proving that the narrow range gives the true breathing rate.

7 Further checks

Looking at individual windows

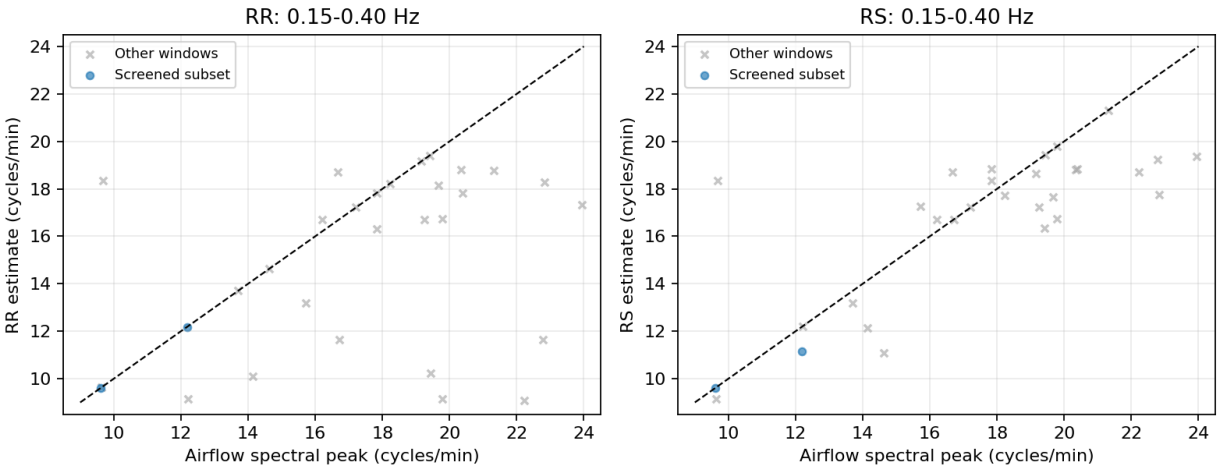


Figure 6: ECG estimates plotted against airflow estimates in the main range. Points on the diagonal agree. The two blue points passed the additional quality screen.

Several RR points lie well below the diagonal, while R–S points show fewer large downward differences. Airflow peaks can also change because of irregular breathing, movement or a different frequency component becoming strongest. Inspecting the waveforms is necessary to understand individual mismatches.

An additional quality screen

A window passed the screen only if it had no flagged RR intervals, usable abdominal data, abdominal and airflow estimates within 2 cycles/min, and neither reference peak at the edge of the search range. The screen did not use ECG-to-airflow error to choose windows.

Only two of 30 windows passed in each range, a retention rate of 6.7%. Missing abdominal samples affected 22 windows and were a major reason for the low retention. In the main range, the two retained RR estimates matched the airflow bins, while R–S MAE was 0.51 cycles/min. These small errors describe too few windows to support a general accuracy claim.

The screening rule therefore needs improvement. Isolated gaps and longer periods of poor signal may need different treatment. Reviewing selected waveforms manually would help develop a more useful rule.

Reading the error measures together

A small bias does not necessarily mean a small error. For example, the broad-range R–S bias was 0.10 cycles/min, but its MAE was 2.33 cycles/min. Positive and negative differences can cancel in the bias. The 30 windows also come from one recording, so they are not 30 independent participants.

8 Comparison using apnea labels

As an extension, the windows were grouped using the supplied minute labels. A window was classed as all N if both minutes had N labels, all A if both had A labels, and mixed if it contained both. The first hour contained six all-N windows, 23 all-A windows and one mixed window.

Table 4: Main-range MAE by minute-label group. MAE is in cycles/min.

Window group	Windows	RR MAE	R–S MAE
All N	6	1.18	1.18
All A	23	3.88	1.97
Mixed A/N	1	0.00	0.51

The methods had similar MAE in the all-N group. R–S amplitude had a lower MAE in the all-A group, where the difference between the methods was more noticeable. The mixed group contains only one window, covering the label change at 13 minutes.

These groups differ in size and occur at different times in the same recording. Their differences may also reflect signal quality or changing ECG and breathing patterns. The results therefore do not show that apnea caused the larger RR errors. Minute labels also do not identify individual breaths or exact event times.

Limits of the analysis

The study has three main limitations. First, it uses only one hour from one person's recording, so the findings may not apply to other people. Second, airflow peaks were used as the reference without manually counting breaths. A peak may represent noise or a repeated pattern that does not match each breath. Third, the supplied beat marks and the simple quality rules can leave errors in the ECG features.

The first window was used during method development, so the study does not have a separate test set. The frequency ranges and two-minute window length are choices that should be tested on further data. The project estimates frequencies; it does not classify apnea or validate a clinical monitor.

9 Learning and problem solving

Connecting data handling with signal analysis

The project required linking several steps that can otherwise seem separate: reading a recording, converting samples to time and voltage, finding beat features, and interpreting a spectrum. Comparing the independently decoded ECG with `wfdb.rdrecord` checked that the calibration had been understood correctly [4, 6]. The calibrated values agreed.

Interpolation was needed because beat times are irregular, while the frequency calculation uses regularly spaced values. Resampling the respiratory signals to the same grid made the comparison consistent. The SciPy documentation helped connect these choices to filtering and periodogram calculation [7, 8].

Decisions based on the data

Table 5: Examples of problems and the responses used in the project.

Problem	Response and evidence
Heartbeat marks were slightly early	Search near each mark for the R peak. The median shift was 40 ms.
An RR interval was unusually long	Keep and flag the 1.37 s interval instead of inserting an unverified beat.
The first result depended on the search range	Compare both ranges across 30 windows. RR MAE changed from 3.21 to 6.41 cycles/min.
The quality screen retained very little data	Report its two-window coverage as well as its errors, and keep all-window results as the main comparison.

Extending the basic method

The R–S feature from Varon and Van Huffel [3] provided a useful comparison with the original RR approach. Applying both methods across the hour, changing the search range and grouping by minute labels added checks beyond the first example. These are established methods, but combining them helped reveal where the results were sensitive to the analysis choices.

The clearest lesson is that an encouraging plot needs further checks. The first RR estimate matched airflow, yet its MAE across the hour was 3.21 cycles/min. Similarly, a screen that leaves only two windows can show small errors while covering very little of the recording. Both examples changed how much confidence could be placed in the initial result.

10 Conclusions and running the code

Conclusion

The project extracted respiratory-related frequency information from RR intervals and R–S amplitude. R–S amplitude agreed more closely with airflow on average in the first hour of record a01. However, errors depended on the frequency range and signal quality. The result supports further investigation, rather than a claim that the method measures breathing accurately in all windows.

The next steps would be to review breaths manually in selected windows, improve the quality checks and test the same settings on additional recordings. Comparing different window lengths would also show whether two minutes is a suitable choice.

Running the analysis

The project folder contains the analysis scripts, data, saved results and the notebook a01_batch.ipynb. From that folder, the numerical analysis and figures can be regenerated with:

Listing 10.1: Commands for the analysis project.

```
python -m pip install -r requirements.txt
python run_analysis.py
python make_report_figures.py
```

The notebook calls the same analysis script. Its stored outputs were captured from a standalone Python execution, rather than a Jupyter-kernel run. The calculations use the supplied recordings, not simulated observations.

Window results are saved in `results/window_results.csv`, with grouped results in summary_metrics.csv. The file `run_info.json` records the run settings and software versions. The default analysis uses the first-hour respiratory extract.

Table 6: Software versions recorded for the analysis.

Software	Version
Python	3.12.14
NumPy	2.3.5
SciPy	1.17.0
WFDB	4.3.1
Matplotlib	3.10.8

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